# *Capturing and revisiting ideas in the design process: A longitudinal technology probe study*

*Peter Dalsgaard, Michael Mose Biskjaer* and *Jonas Frich*, Aarhus University, Helsingforsgade 14, Aarhus, Denmark

*Managing ideas is a central activity of professional designers and an essential element in the creative design process. However, while designers use a diverse range of tools to capture, manage, and recollect ideas, they often lack well-established and proven efficient routines for doing so. To better understand how these idea-centered design processes play out and, potentially, might be improved, we conducted a longitudinal study with ten professional designers where we deployed a technology probe in the form of an app, Recollect, to explore critical aspects of their idea management practices. Based on weekly interviews, we contribute four central insights, which pairwise form two current tensions related to idea capture and idea recollection among designers. These two tensions are that 1) designers have similar conceptions of what constitutes a good design idea, but have very diverse practices for working with ideas; and that 2) designers find value in revisiting ideas, but lack structured and efficient approaches for doing so. We discuss the implications of these findings, and we propose that the complexity of designers' idea management practices call for more research.*

© *2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).*

*Keywords: creativity, human–computer interaction, design tools, design process(es), design studies*

Ideas are central to creativity (Smith & Ward, 2012), and many activities in creative work revolve around the generation and transformation of ideas to achieve innovative outcomes (Chulvi, Gonźalez-Cruz, Mulet, & Aguilar-Zambrano, 2013). This central position for ideas also goes for design, as designers rely on extant knowledge and repertoires (Finke, Ward, & Smith, 1992) and search for inspirational stimuli (Goncalves, Cardoso, & Badke-Schaub, 2014) to develop, combine, and transform ideas to reach novel and valuable solutions. For this reason, the process of ideation has long been a key topic in design research (see e.g. Davies & J Talbot, 1987), but it has proved challenging to examine ideation mechanisms empirically over time, especially in vivo. This means that there is a lack of tool-focused studies of longitudinal idea development processes compared to singular

**Corresponding author:**
Peter Dalsgaard
dalsgaard@cavi.au.dk

www.elsevier.com/locate/destud
0142-694X *Design Studies* **88** (2023) 101200
https://doi.org/10.1016/j.destud.2023.101200
© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

ideation sessions (Vargas Hernandez, Shah, & Smith, 2010). This limited body of research might seem paradoxical insofar as designers make frequent use of tools and resources while working with ideas (Inie & Dalsgaard, 2020).

Coughlan and Johnson coined the term *idea management* (2008) to describe the creative practices that designers and other creative practitioners engage in to capture and keep track of their creative ideas. The authors identified three core purposes of idea management: a) retention and organization of ideas; b) feedback, evaluation, and development of ideas; and c) communication and collaboration around ideas (Coughlan & Johnson, 2008). Current knowledge is also limited regarding how designers support their idea management by using *tools and resources*. Studies show that designers capture these ideas using a diverse range of tools and resources (Coughlan & Johnson, 2008), and that they find value in revisiting prior creative ideas in their design processes. Even so, only a minority of designers seem to have well-established routines and workflows for capturing and revisiting ideas (Coughlan & Johnson, 2008). The absence of integrated and efficient modes of working with ideas may entail that some innovative ideas will be ignored or completely forgotten in a design process.

To examine these critical aspects of design idea management, we present a longitudinal, exploratory in-vivo study of an idea management technology probe, *Recollect*, which is a lightweight smartphone application. The Recollect application has been designed to support designers' 1) *rapid capture of ideas* and 2) *subsequent revisiting of these ideas* corresponding to two of the key purposes identified in existing literature (Coughlan & Johnson, 2008). The present study design was chosen because the process of generating and developing ideas in design often unfolds over longer periods of time and in unpredictable, co-evolutionary problem-solution patterns (Dorst & Cross, 2001) that are difficult to study in discrete, controlled experiments. The Recollect technology probe was used and evaluated by ten professional designers within the domains of graphics, digital products, and user experience for approximately one month to explore the following two open-ended research questions:

*RQ1*: *How do professional designers collect and revisit design ideas?*

*RQ2*: *What are the implications of introducing digital tools for idea management in professional designers' creative processes*?

Given the study's exploratory nature, the Recollect technology probe (Hutchinson et al., 2003) was introduced into the everyday professional practices of the ten interaction designers to gain insight into their idea workflows and practices; particularly how they would capture, manage, develop, and share their design ideas. Based on 40 qualitative interviews conducted during

and immediately after the probe study, we used thematic analysis to illustrate how Recollect impacted the professional designers' perceptions and practices of idea management. The study offers insight into *digitally supported idea management processes among professional designers* and contributes two main insights, namely that 1) *the designers have similar conceptions of what constitutes a good design idea, but have very diverse practices for working with ideas*; and that 2) *designers find value in revisiting ideas, but lack structured approaches for doing so*. The study should be of relevance to researchers and practitioners with an interest in design processes in general and creativity support tools and idea management in particular.

# 1 Idea management in design

## 1.1 The prominent role of ideas in design creativity

The concept of creativity revolves around the generation and exploration of ideas (Guilford, 1956; Smith & Ward, 2012). As a key concept in creativity research since the 1950s, divergent thinking has often been measured by the quantity and attributes of the ideas produced (Kaufman, Plucker, & Baer, 2008). Creativity is challenging to pin down since proposed definitions vary and tend to be domain-specific (Puryear & Lamb, 2020). According to one of the most influential recent suggestions, creativity is "the interaction among *aptitude*, *process*, and *environment* by which an individual or group produces a *perceptible product* that is both novel *and useful* as defined within a *social context*" (Plucker, Beghetto, & Gayle, 2004, p. 90, orig. emphasis). Construing creativity in this way suggests that ideas are assigned an intermediary role in creative processes, as they, through refinement and transformation, can be materialized or instantiated as a ''perceptible product' of some sort. M Amabile (1988) similarly placed the development of ideas at the centre of creative processes, stating that "*creativity is the production of novel and useful ideas by an individual or small group of individuals working together*" (p126, orig. emphasis). According to Christiaans (2002) creativity is a key criterion of design, and Haugen Askland, Ostwald, and Williams (2010) discerned two prevailing approaches to creativity within various design disciplines. The present design study is informed by this understanding of the importance of ideas at various medial stages in the creative design process in the sense that "the function of inspiration within the context of creativity is the actualization of creative ideas into creative products" (Oleynick, Thrash, LeFew, Moldovan, & Kieffaber, 2014, p. 8).

Although the relevance of ideas in design creativity is well-established, the difficult ontological question of what constitutes a creative idea remains. Following the above definition by Plucker et al. (2004), the given context frames what various stakeholders might refer to as a design idea. Novel solutions often emerge through a combination of two or more ideas in ways not

immediately obvious, see (Arthur Koestler, 2020). These ideas, then, are effectively the 'raw materials' from which the final design solutions take shape. For an architect, such a design idea might be a combination of geometrical shapes that express harmony in a building, a specific inflow of light that creates a desirable ambience in a room, or a new type of cladding material. Similarly, a graphic designer might appreciate a new style of typography, a unique combination of contrasting colours, or an unexpected perspective in a photograph, while a UX designer might count a design idea as a simple but efficient change to an app interface to ensure a smoother user journey. As such basic examples suggest, design ideas always depend on context and perspective. Other stakeholders, e.g., a developer or a resident in the first example or users, clients, or a company in the other two, will inevitably have another perspective on what constitutes a design idea. This points to the need to examine ideas in design creativity in more detail, not least how such ideas emerge and evolve over time and how designers capture and recollect them.

### *1.2 Time, tools, and materials in idea management*

There are multiple ways of moving through creative processes which are typically depicted as a sequence of phases based on the development of novel and useful ideas. M Amabile, Conti, Coon, Lazenby, and Herron (1996) characterized creative processes as composed of four activities: a) problem or task identification, b) preparation, c) response generation, and d) response validation and communication. The preparation phase [ad b)] and the response generation phase [ad c)] are especially relevant to the present study, as they pertain to the "building up and/or reactivating store of relevant information and response algorithms" and " "search memory and immediate environment to generate response possibility" (1996, p. 113). This resonates with phase-based models of the creative interaction design process such as the one presented by Preece, Rogers and Sharp (2002), which is based on four basic phases, and Löwgren and Stolterman's (2004) description of the iterative progression from early visions to detailed specifications. Howard, Culley, and Dekoninck (2008) identified six common phases: a) establishing a need, b) analysis of task, c) conceptual design, d) embodied design, e) detailed design, and f) implementation. Ideation in the early phases of design if often emphasized in these and other phase-based models, and many methods have therefore been developed to spur design ideation (e.g. Biskjaer, Dalsgaard, & Kim, 2017; Casakin & Levy, 2020; Vargas Hernandez et al., 2010).

Creativity research has shown that the development of new solutions and the emergence of insight or eureka moments usually require incubation (Cardoso and Badke-Schaub, 2009; Tsenn, Atilola, McAdams, & Linsey, 2014). These findings entail that deliberately putting a problem aside for some time only to return to it later (either spontaneously or through planning) can, paradoxically, enhance creative problem solving. A meta-analysis by Na Sio and

Ormerod (2009) found support for the existence of incubation effects and identified a range of potential moderating factors such as task and duration of incubation. In relation to the creativity of designers, it is especially noticeable that when engaged in "creative problems that require a wide search of knowledge, individuals benefit from an incubation period" (Na Sio & Ormerod, 2009). Designers often develop many ideas, of which only a few makes it to the latter stages of a design process. Ideas that make it to these latter stages of projects are often transformed and developed iteratively (Catmull & Wallace, 2014; Kocienda, 2018), and can evolve from simple sparks or idea fragments to more complex assemblages, in which multiple ideas are combined (Sosa, 2019). Moreover, as demonstrated in literature on the co-evolution of solution and the problem spaces, the criteria by which ideas are evaluated and developed also change throughout the process as designers develop a stronger understanding of the problems they address and the potential solutions to these problems (Finke et al., 1992; Lou Maher & Poon, 1996).

To be able to capture ideas and store them for future recollection, for example to reap the benefit of incubation, creative professionals such as designers rely on tools and materials to support their memory as the basic internalization of ideas. Research into distributed cognition (Hollan, Hutchins, & Kirsh, 2000; Howard et al., 2008) and 4E Cognition (i.e., embodied, embedded, enacted, and extended cognition) (Newen, De Bruin, & Gallagher, 2018) has shown that using tools and materials augments the designers' cognitive capabilities by improving the designers' ability to recall, manipulate, and communicate creative ideas. Research into spaced repetition has demonstrated that retention and recall of prior knowledge, including ideas, can be enhanced by revisiting ideas at certain intervals (Ausubel & Mohamed, 1965). Even so, recent studies suggest that designers rarely revisit old ideas and sources of inspiration, even if they spend time and effort to collect them (Inie & Dalsgaard, 2017a, 2017b). This lends support to the assumption behind the development of the Recollect technology probe, namely that better means for capturing and revisiting ideas can benefit designers, since it may help them to better retain and activate creative ideas throughout and after the design process itself. This makes it relevant to examine the current body of research on idea management.

In design research in general, and in interaction design in particular, recent research has contributed with studies on how creative professionals manage their ideas (Tim Coughlan & Johnson, 2008; Sharmin et al., 2009). Most closely related to our focus in this paper are Inie et al.'s studies of archetypal idea management approaches of designers in practice (2018), and frameworks of current strategies and tools for idea management (2020). Coughlan & Jonhson (2008) found that most designers implicitly or explicitly develop workflows for idea management and often build up idea repositories in the form of e.g. analog notebooks or digital collections of inspirational material and examples from past projects. These systems and workflows support

designers in capturing, developing, organizing, retrieving, and sharing ideas (Inie and Dalsgaard, 2020). Both Inie and Dalsgaard (2017a, 2017b) and Coughland & Johnson (2008) found that professional practices around idea management are diverse and idiosyncratic, stressing the need for further studies on how ideas in design practice are captured, managed, and exploited in order to identify patterns and develop a more comprehensive understanding of the phenomenon. This body of research shows that although the tools and systems deployed to support these idea management practices seem pivotal in design, this creative practice has not yet been thoroughly examined. Similar studies of idea management have subsequently been carried out to examine how scientists manage ideas (Inie, Frich, & Dalsgaard, 2022). Of further interest for our study of Recollect is Inie & Dalsgaard's typology of design ideas (2017a, 2017b), since it describes the different types of ideas that designers commonly work with. Surveying the literature on design creativity and ideation, the authors identify four main types of design ideas pertaining to a) framing the problem, b) opportunity, c) suggestion for part-solutions, or d) suggestion for solutions. We will employ this typology to examine and discuss the types of ideas present in our study of Recollect.

### *1.3 Summary: the need for longitudinal in-vivo studies of idea management in design*

The concepts of ideas and creativity are closely interlinked and play a central role in design. Designers often use external resources and tools to capture, articulate, develop, and manage ideas as part of their creative process. Research (Inie & Dalsgaard, 2020) recently indicate that designers——despite having developed unique practices around the management of ideas by piecing together tools and approaches——rarely *revisit* their creative ideas.

This points to an untapped potential for better supporting designers in both managing and revisiting ideas. While it is unclear what impact the infrequent, or even absent, revisiting of ideas has on the designers' general professional ability and creative behaviour, the importance of incubation time for creativity is established (Na Sio & Ormerod, 2009). In addition, there are limited longitudinal in-vivo studies of design ideas to show how ideas emerge and evolve in practice (see e.g., Ball & Christensen, 2018). These contributions inform the present study in which we adopt a longitudinal in-vivo approach to explore the potentials and limitations of employing a custom-designed digital application, the Recollect technology probe, for professional designers to better retain and (re)activate their design ideas over time. Finally, the study focuses on domains of design (UX, product, graphic, etc.) where digital representation forms are assumed to be more likely than e.g., fashion or furniture design (although serious digitalization are also expected to impact these.)

# 2 Method

## 2.1 Study design: technology probes, interviews, and thematic analysis

Our study is based on the deployment of a *technology probe* (Hutchinson et al., 2003) in the form of the *Recollect* mobile web application. The technology probe approach is characterized by developing a simple, but fully functional product and then deploying and studying it over time in participants' real-life use context (Hutchinson et al., 2003). The goal of the technology probe approach is not to gather unbiased data about the use of the product, but to study changes in, and reflections on, participants' routines and practices over time in relation to the probe. For this reason, the Recollect probe was deliberately designed with few, but fully functioning features (described in more detail in section 3.3) to allow for users' appropriations and reactions to thereby inform further research and development. This allow us to directly answer RQ2: *What are the implications of introducing digital tools for idea management in professional designers' creative processes?* and acts as an impetus for approximating RQ1: *How do professional designers collect and revisit design ideas?*

This study design was chosen, since it a) offers a high degree of ecological validity, and b) permits a focus on a particular aspect of practice, in this case the capture, management, and development of ideas over time. As argued by Christensen and Ball (2014), design studies are traditionally conducted in contrived settings, which strengthens internal validity at the expense of (lower) ecological validity. In essence, this means ignoring the messy nature of real-world professional design where collaboration, expertise, and organizational structures all play non-negligible roles. The present study is somewhat aligned with Christensen and Ball (2014) who proposed an in-vivo protocol analysis approach. However, since one of our main concerns was getting access to an understudied domain, namely professional designers' longitudinal idea management practices in relation to technology, we opted for an alternative to protocol analysis. Our approach is akin to a field-experiment (McGrath, 1995); an approach in which the researcher seeks to gain more precise information about certain aspects by manipulating one feature and giving up some of the unobtrusiveness of a traditional field study. In our case, this means manipulating a particular aspect of designers' idea management practices via the Recollect technology probe to gain insight into the potentials and limitations of digital tools to support these creative design practices. This is also the reason why we opted for a single-user technology probe rather than a multi-user system, which in addition to introducing more variables would be significantly more complex and resource-demanding to develop.

Consequently, the goal of this study was not to evaluate whether Recollect would be superior to participants' existing practices or tools, nor to assess

whether the participants 'liked' Recollect or "would recommend it to others" (Hutchinson et al., 2003). Rather, the Recollect technology probe served as a *catalyst* for generating insights into particular challenges, perspectives, and experiences pertaining to the recollection and management of creative ideas through the use of digital tools.

## 2.2 Study procedure: longitudinal study with recurring interviews

Ten experienced designers (7M, 3F, see Table 1) were recruited, each with three to fifteen years of relevant professional experience since obtaining a design-related university degree. Recruitment took place via invitations posted in professional design groups on LinkedIn and Facebook. The participants comprised a mix of consultants and in-house designers. Given their work experience, all participants had developed and managed ideas in several design projects prior to participating in the study. Participation in the study was voluntary and uncompensated. The participants stated that their incentives for participating were the opportunities that came from experimenting with the Recollect mobile web application, reflecting on their own practices, and gaining new insight into idea management.

### 2.2.1 Interviews

To account for the potentially slow change of working habits, the study was conducted over the course of one month. Short weekly interviews and a longer final interview were conducted by all the authors following a semi-structured approach (see Table 2). The interview-guide for the short interviews included

**Table 1 List of participants' titles and experience**

| *Participant* | *Experience* | *Title* |
|---|---|---|
| P1 | 15 years | Senior Service Designer |
| P2 | 13 years | Graphic Designer |
| P3 | 6 years | Product Specialist |
| P4 | 2 years | User Experience Designer |
| P5 | 6 years | User Experience Designer |
| P6 | 4 years | Senior Product Designer |
| P7 | 8 years | Senior Producer |
| P8 | 7 years | Customer Experience Designer |
| P9 | 5 years | Senior User Experience Designer |
| P10 | 3 years | User Experience Designer |

**Table 2 Overview of study procedure**

| | *1 Month* | | | |
|---|---|---|---|---|
| W0 | **W1** | **W2** | **W3** | **W4** |
| Questionnaire + App setup | Short interview | Short interview | Short interview | Longer interview |
| | ~13.5 min | | | ~35 min |

six questions with potential follow-up questions. The questions related to the following subjects.

1. When and in what context the interviewee used Recollect for the week
2. Examples of how and if Recollect impacted any current projects
3. How and if Recollect impacted other work
4. Cased where revisiting ideas in Recollect was especially influential
5. General perception of Recollect
6. General comments about the previous week incl. feedback etc. Before initiating the interviews, a survey on participants' experience with and attitude toward idea management inspired by Inie and Dalsgaard (2017a, 2017b) was conducted. The participants were instructed to use the Recollect probe to capture ideas, hunches, and sources of inspiration for their creative design work, and they were encouraged to spend at least two-three minutes per day browsing through ideas. The weekly interviews were 13.5 min on average, while the longer exit interviews were 35 min on average. All interviews were conducted online as a video conference call or via phone. The interview data thus amounted to approximately 10 h in total. In the trade-off between longitudinal aspects as opposed to larger a sample size we emphasized the former.

## 2.3 Technology probe: the Recollect mobile web application

The Recollect technology probe was designed as a mobile web application intended to be used on a modern smartphone, which all participants had and would use extensively during their workdays. Recollect was designed and built in collaboration with professional developers at the authors' research lab [omitted for anonymity]. At its core, Recollect offers functionality for capturing, revisiting, sorting, and annotating ideas in a straightforward and lightweight manner.

### 2.3.1 Description of functionality

The functionality of the Recollect web application was deliberately sparse, as suggested by the technology probe approach. Users could log on to the app individually using their credentials from existing web-based services such as Google, Microsoft, Linkedin, Twitter, etc. Once logged on, a single, randomly selected idea from the user's collection was immediately presented. The user could then swipe up and down to randomly move between ideas (Figure 1, screen 3). By swiping left or right on an idea, the user could mark or unmark the idea as a favourite (Figure 1, screen 2).

In addition to pure text, ideas could include images or video-clips alongside text (Figure 2, screen 1); specific collections, to which ideas could be sorted, selected, and created from the menu (Figure 2, screen 2). New ideas could

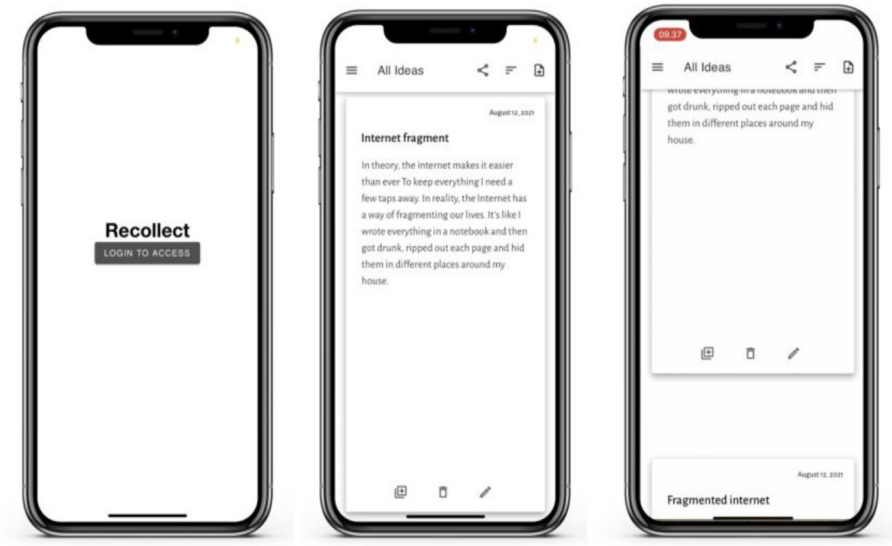



*Figure 1 Screenshots of the Recollect app: Login, a text note, and navigation between notes via vertical scrolling*

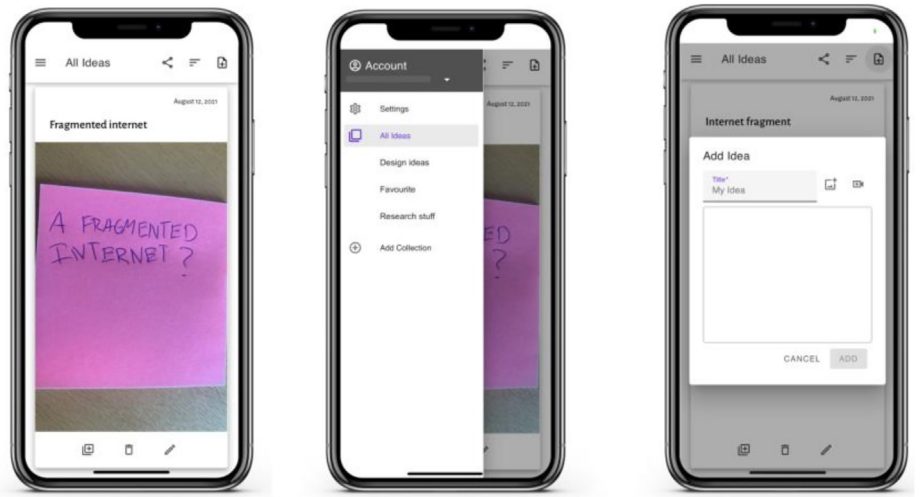



*Figure 2 Screenshots of the Recollect app: An image note, menu bar with collections of ideas, and adding a new idea*

easily be added (fig. screen 3), and existing ideas could be annotated, deleted, and sorted.

## 2.4 Thematic analysis

To analyse the qualitative interview data, we employed thematic analysis, which is a well-established interpretivist methodology originating from psychology but now widely appropriated across academic disciplines (Braun &

Clarke, 2006). Thematic analysis involves grouping the qualitative data into codes and then coherent themes to describe the data in more detail (Braun & Clarke, 2006). This often involves distinguishing between latent and semantic interpretations where the researcher will often move from semantic to latent as the analysis progresses (Braun & Clarke, 2006). Concretely, all interviews in addition to supplementary notes taken during the interviews were coded individually by the authors. During this process, joint meetings were held frequently among the authors to share insights. The authors then collectively conducted two axial coding sessions. Finally, a collective coding session was carried out, resulting in the themes presented in the following section. It should be noted that the authors' prior knowledge about idea management and design may have entailed a tacit framing of the interviews and subsequent coding. Based on the thematic analysis we present six main themes. These themes were selected based on their prominence in the interviews and their relevance to creativity and idea management in order to best address the research questions. This means that themes pertaining to e.g. participants' assessment of technical and usability-related issues with Recollect were side-lined, as long as they did not directly influence participants' use of the technology probe. This approach is aligned with Braun and Clarke (2006) who advocated for focused presentations of thematic analyses based on themes' centrality and relevance to the research questions.

## 3 Results

This section presents six main themes that resulted from our thematic analysis. The themes are interrelated, and we have chosen to present them sequentially on the basis of how they typically occur chronologically for participants, such that the first themes mainly address aspects of idea collection when participants first encounter or develop ideas, with the latter themes moving into how participants subsequently manage, revisit, and further develop or refine ideas. This serves as the basis for the subsequent discussion, in which we examine the relations between the themes and discuss them in light of related work on idea management.

### 3.1 Ideas were collected in diverse ways within and between projects

The study participants described that the primary goal of capturing ideas was to avoid forgetting them and to put them aside temporarily in order to focus mental efforts on other tasks at hand. P5 expressed this as "getting it off my chest." There was consensus among participants that capturing ideas was valuable and worthwhile, providing strong intrinsic motivation for using Recollect in conjunction with already-established idea collection routines.

The interviews revealed much diversity in when and under which circumstances the participants collected ideas. In some circumstances, ideas were

collected in designated product research and idea development sessions; for instance, in meetings with clients to develop specific solutions. This was prominent in the case of P4, who would most often capture ideas of product solutions sketched out on whiteboards or sticky notes in joint meetings with clients or collaborators. P10 would usually collect ideas as annotated screenshots via her phone in the process of doing research for new projects by examining existing product solutions.

But just as often, participants reported collecting ideas ad hoc or in between events and projects. P6 explained that ideas would typically emerge in between events such as when going from one meeting to the other. He therefore appreciated an always-at-hand tool for idea collection such as Recollect. P2 reported on adding ideas to Recollect during breaks because the ideas might become relevant for future projects, and P1 mentioned using Recollect for collecting sudden insights while commuting by bike or on foot to and from work.

Several participants also reported on cases in which circumstances impeded or were not conducive to idea generation and collection. As a case in point, P4 stated that in some circumstances it seemed untimely to collect ideas in meetings, which he mainly did via photos, because it would be socially inappropriate or break the flow of a creative process. In these cases, he would try to process and collect ideas after these sessions. P2 described in one interview that his team was currently in a phase where they were "not allowed to generate new ideas" because they had to focus on already-proposed solutions. We encountered this views on 'phases' in several interviews. The study participants reported idea development as being crucial in some phases of design projects, while being unimportant or even obstructive in other phases. In fact, P5 stated during one interview that they, at that specific stage, were simply not in the phase of the design process where it would make sense to capture ideas. Similarly, P4 reported on having a focus on "only capturing ideas that were relevant in the current project."

### *3.2 Ideas were collected in a variety of formats*

While the Recollect web application supports idea capture in a range of formats such as links, text, video, audio, and images, the study participants had a strong preference for capturing ideas in a visual format, mainly as images, which they would often annotate with additional information in writing. This was evident in various ways. Most of the ideas captured by P10 were screenshots taken with the phone camera with added text to explain and contextualize the idea. P4's ideas were predominantly photos of whiteboards and physical notebook pages with sketches of ideas while P7's were mainly photos of sticky notes with ideas. It was clear that ideas were not always "born" in the Recollect mobile web application. Rather, they were externalized in different formats such as whiteboards, sticky notes, and screenshots

first and then subsequently captured in the system. Moreover, participants interpreted and appropriated Recollect in response to their existing workflows for idea management and development.

The ideas collected were predominantly solution-oriented, either in the form of ideas that could directly inform ongoing projects or in the shape of inspirational material that could inform solutions in future projects. An example of a solution-oriented idea can be found in P4 making a systems flowchart of how to implement a product feature in a notebook and capturing it as a photo in Recollect; and an example of inspirational material can be found in P10 capturing a screenshot of from an online marketing campaign and annotating elements that she thought could inspire potential future campaigns in her own organization. In section 5.1, we will further discuss the solution-oriented focus on ideas considering related research.

Even though all participants had similar job functions, educational backgrounds, and technical expertise, they had very different perspectives on and routines for idea management and development. Their use of Recollect reflected this diversity. One example is P6's "bullet journal" (Carroll, 2018) as a pivotal tool to organize and manage his tasks and ideas, so he set up routines for processing and manually synchronizing ideas between Recollect and his paper-based bullet journal. P7 had a routine of using sticky notes extensively for articulating and developing ideas, so his use of Recollect was centred around capturing photos of these sticky notes to store and revisit them. Finally, P4's had established routines for idea development that relied on the use of whiteboards and sketches in joint meetings, which were then captured and consolidated in Recollect in a way like that of P7.

### *3.3 Ideas were recalled and revisited infrequently*

Recollect was designed specifically with the goal of inviting users to actively revisit ideas. The study participants were prompted to quickly assess the ideas in their archive using the aforementioned swipe gesture, with the additional option of adding notes and new ideas. On the basis of the qualitative data, two themes were conceptualized in relation to the activity of revisiting ideas: participants' perceived barriers to recollection and how ideas were recalled and revisited.

One of the focal points of the interviews was how participants would go about revisiting ideas, and how the Recollect web application played a role in doing so. P1 would set up time during the evening when her children were asleep to go through her ideas from the day. This approach, however, was interrupted during the period when the study took place, and the change of routine posed challenges to her revisiting ideas:

> "Yes, I've changed my daily routines.. or maybe it's not me, but my kids … I usually do it in the evening, just after my kids go to sleep.. and at the moment one of them doesn't want to go to sleep, so I don't have as much time in the evenings as I used to, so I have to remember to do it at my workplace … so I think I remembered to do it a couple of times, but yeah … I need to find a new time slot, where I can do it …" (P1).

Another participant, P6, echoed the approach taken by P1 (at least before the routine was interrupted), mentioning how he would revisits his bullet journal on a monthly basis. He explained how it was an integrated routine, and that now, as part of using Recollect, he was syncing the content between the two media. Beyond these individual practices around revisiting ideas, P3 had routines in place that would allow for some incubation time. P3 would achieve this by circulating ideas among peers. He explained how he would normally go about doing this: "What I often do, when I make these things [ideas] is that I send them to my teammate [ …] It works quite manually where I would find them [ideas] in my camera roll". Likewise, P10 similarly stated that they "need to circulate ideas and put them to work, e.g., in Slack. Otherwise they just wither and die. Alternatively, I need to prepare ideas in advance of tasks or projects that I know are coming up." This practice of circulating ideas was construed thematically as a way of 'revisiting' the ideas, which would potentially be an important feature for future development of digital idea management tools. The importance of circulating ideas captured in the app and "putting ideas to work" by sharing them among colleagues in order to get their thoughts and feedback was especially pronounced by participants P10 and P3, who found this practice of ongoing idea distribution critical in their idea development process.

### *3.4 Differing perceptions of what an idea management system could and should do presented barriers to idea recollection*

Although revisiting ideas was a pivotal aspect in the study, participants rarely mentioned revisiting previously-collected ideas in Recollect. This is noteworthy because Recollect was developed to support and investigate this exact practice. For some, Recollect was treated somewhat similar to a to-do list. P8, for example, voiced how he saw Recollect more akin to a journal or a to-do list: "I've used it in a very functional manner. In my head, it's been to-do lists and tasks. Maybe the function should rather be to gather and collect and then before I go to bed I'm just able to get rid of some thoughts." Researcher: "Are you thinking as a way to gather ideas which you can then look at it at night or once a week or similar?" P8: "Yeah I'm thinking more like today when I go to bed, is there something I need to remember to do tomorrow, maybe I should remember to send that e-mail. Could be both major and minor things." On a similar note, it did not make sense for P4 to revisit ideas. He stated this in the exit interview: "What Recollect is missing now is a resurfacing feature, because

there is no resurfacing, it's a stack of things that I then have to swipe through, so nothing is coming to the surface by itself, it's something that I'll have to bring to the surface." In a prior interview, P4 had also explained that the way ideas were visualized in Recollect were insufficient for him when dealing with existing ideas. Specifically, he expressed a wish for a more classic "overview" of several ideas at the same time: "If you could combine it with an overview, you know, try to visualize to me that here is a giant pile of great ideas in order for me to actually want to dive in. I think that would really help."

### *3.5 Participants sought to embed Recollect into existing idiosyncratic creative workflows*

Since all the participants were trained designers and had similar educational backgrounds, their awareness of the importance of design phases and the need to establish routines and workflows around their creative processes was to be expected. Still, the study revealed some distinct differences between their use of the application in relation to how ideas were developed.

In terms of design phases, the Recollect web application was often deployed as a preparatory tool to enable participants to consolidate ideas from preliminary phases in the design process before moving on to the next phases. P4 and P10 reported how they found Recollect expedient for preparing ideas for subsequent use even though the ideas were not directly actionable in the given situation. P7 explained that he found it important to be able to collect "all crude ideas in one place, so that you always know where to look to." This meant that Recollect would be useful as a tool that could "bridge the ideation phase and the design phase." According to P7, very few existing support tools serve this particular function as a transitory step between phases in his already well-established creative workflow.

Another critical aspect to several participants was the distinction between *refining* and *remembering ideas*. P8 was particularly reflective about the dissimilarity between these two ways of using Recollect. Similarly, P7 used an analogy to explain the difference——"a waterfall" versus "a diamond." The waterfall metaphor referred to the practice of continuously capturing a "cascade" of ideas by getting them into the Recollect app in order to remember them without further discernment. P8 would like to tick off ideas as a to-do list as soon as he could complete the idea by "pushing it" onto a concrete project. This approach resembled participant P7's approach of digitizing a sticky note containing an idea in order to remember it before using the idea as input for developing slides to be presented to his group members. Conversely, the diamond metaphor introduced by P7 referred to the practice of iteratively curating and refining a given idea in Recollect until it had reached a more mature and advanced stage in the development. This specific practice was underlined by P6, who reported how he would typically add a crude idea to

the app and then continue to refine it by expanding on or editing notes, and adding supplementary imagery within the app. He would continue this process until the idea was so crystal clear, as it were, that he found it solid enough to take to the next step. This involved transferring the idea to his computer in order to further develop, draft and design the idea using professional design software tool (Figma in this case of P6).

### *3.6 Recollect prompted reflections on what the ideal idea management tool could be*

While all study participants expressed that Recollect was relevant in their workflows, they also voiced concerns about whether it could fit into their current, well-established practices of working with ideas. In some cases, this led to reflections on what an ideal idea management tool could be. To our knowledge, forming a theme based on ad hoc speculations on potential features is atypical in a thematic analysis. Nevertheless, we chose to represent it in the form of theme since it has its own distinct organising concept (Terry, Hayfield, Clarke, & Braun, 2017) (e.g.: "what could be"), and as it may be in part due to the nature of our participant sample (UX, product, graphic designers etc.)

P7 explained that Recollect might serve as a "documentation hub" or "idea collection bank; " however, in its current state, the app's "clunky" appearance and limited usability meant that it introduced one or more "extra steps" in the creative idea development process without these superfluous steps adding any immediate value. Other participants' impressions of Recollect focused more on missing features for the application to be genuinely useful in existing professional workflows. As a case in point, P8 expressed how he missed a completion feature to indicate that an idea had left the preliminary stage and was now ready for further creative development, either by him or his team members or other collaborators. Another issue that emerged concerned the lack of an overview of all ideas in the app. One participant (P4) reported how he needed a complementary desktop application for Recollect to be sufficiently useful. As the number of ideas was bound to grow over time, he found it critical to be able to lay out an idea more comprehensively in a way that would enable him to select and work on the most relevant ideas for further refinement. P8 echoed this feature-oriented point by expressing two concerns: Firstly, that the Recollect web application needed a "canvas-like" function where all notes could be "spread or fanned out" in order to let the user get an overview. Secondly, that the application lacked a functionality that would enable the user to distinguish between "the original idea and what had been added later." Adding to these concerns was the lack of a full-application search function that would provide a quick and easy overview of the different ideas contained within Recollect.

# 4 Discussion

In light of prior research into design ideas and creative workflows in design practice, four key findings stand out from our study. Pairwise, they form two tensions, which we will discuss in the following. The first tension, here called 1), is that 1a) designers have similar conceptions of what constitutes a good design idea, 1b) but they have very diverse practices for working with ideas. The second tension is that 2a) designers find value in revisiting ideas, but 2b) they lack structured approaches for doing so.

## 4.1 Designers have similar conceptions of what constitutes a good design idea, but they have very diverse practices for working with ideas

As detailed in the two first themes (section 4.1 and 4.2) **participants first and foremost conceived of design ideas as** *suggestions for solutions* **or** *suggestions for part-solutions*. This was a somewhat surprising finding given the more diverse conceptualizations of design ideas in prior research (Inie, Dow, & Dalsgaard, 2018), but may result from the genres of design represented in our sample (graphics, digital products, and user experience). Prior research into the notion of design ideas has examined a range of conceptualizations of what an idea could be. The typology of design ideas presented by Inie and Dalsgaard (2017a, 2017b) summarized this research and distinguished between design ideas pertaining to *problem framing*, *opportunities and inspiration*, *suggestions for solutions*, *suggestions for part-solutions*, *design moves*, *insight moments*, or *plans for action*. Informed by this, we invited participants to think broadly about the notion of ideas during their introduction of the Recollect technology probe and in the subsequent interviews. As an example, the written instructions for the study proposed that participants could use Recollect to capture and revisit "ideas, hunches, and sources of inspiration for your work," prompting participants to cast a wide net when collecting ideas.

The solution-oriented focus may in part explain why the participants gathered a relatively small number of ideas: if they mainly conceive of ideas worth capturing as solutions or part-solutions to current or near-future design problems and omit other types of ideas, e.g., hunches, potential sources of inspiration, or ways of reframing problems, this would strongly limit the scope of what is collected.

The solution-oriented focus has clear implications for the ideas that participants collected, developed, and revisited. In terms of idea collection, it meant that they primarily collected content that they could imagine would contribute to solving a design problem in a current or planned project. Once an idea had emerged and been recorded, the participants focused on how that idea could be implemented and executed, rather than e.g., how one or more novel sources of inspiration could offer original concepts for new projects or alternative

framings of existing ones. In some cases, participants collected ideas for projects they knew to be on the horizon, but again mainly with a solution-oriented perspective. In terms of revisiting ideas, most participants did not have any strategies or routines in place for actively revisiting previously captured ideas beyond the scope of ongoing projects. Again, this may in part be explained by the fact their focus was in most cases on ideas that could be implemented in the very near future. Recollection of older ideas was not actively pursued, although participants reported on sometimes stumbling on old ideas, e.g., when leafing through notebooks.

While they had a largely similar conception of design ideas as suggestions for solutions or part-solutions, **participants had very different approaches to what may be labelled 'idea workflows'**; i.e., the work practices and routines specifically pertaining to collecting, managing, revisiting, sharing, and developing ideas. Given that the participants had similar job profiles, educational backgrounds and conceptions of design ideas, the diversity of their idea workflows was noteworthy. These differences concern the circumstances under which they would capture ideas, which analogue and digital media and systems they would employ, how they would manage and work with them subsequently, whether to share them with others and if so, how. Some participants would capture ad-hoc ideas as photos on mobile devices and then share them as soon as possible with their team with a minimum amount of annotations or comments, while others would jot down ideas in physical notebooks during client meetings and then transfer and develop the ideas individually using their desktop computer setups, preferring to arrive at a well-developed concept before communicating it to others.

We found that participants had previously appropriated and pieced together tools that were generic, developed for other purposes or which only addressed subsets of participants' idea workflows. This is unsurprising, as there are few, if any, digital tools developed specifically for the purpose of supporting idea workflows. Participants expressed that they would appreciate digital systems that could support them in better structuring and managing their idea workflows, but they did not know of tools that could directly help them accomplish this. Despite our research into this field, we could not point the participants to dedicated tools, since this is a category of specialized software that is largely lacking. On the one hand, there seems to be a great potential in developing systems for better supporting idea management and idea workflows in design. On the other, given the significant differences in participants' idea workflows, it could very well be that the aspiration to develop a one-size-fits-all tool for idea workflows is unlikely to succeed. A more fruitful approach could lie in developing systems that can be finely customized to designers' existing workflows and configurations or tools, or which are modular and can be combined by designers and other knowledge workers as they develop their idea workflows over the course of time.

### *4.2 Designers find value in revisiting ideas, but they lack structured approaches for doing so*

The study participants all expressed over the course of the study that **revisiting design ideas was valuable** to them, especially because it helped them remember and reflect on potential solutions. Moreover, they found value in having a repository in which they could quickly capture ideas, and then return to them later when they could dedicate time to consider the ideas more carefully. This finding is in line with prior research (Inie et al., 2018) which found that resurfacing ideas can prompt reflection on the idea itself and on the designers' own practices and perceptions as well as motivate designers to work on ideas they had forgotten. When taking into consideration how valuable it can be for designers to revisit ideas, a noteworthy finding from our pre-study survey was that **participants self-reported that they had unstructured approaches to working with ideas**, including revisiting ideas. This was true even for participants who otherwise reported that they generally had a very structured approach to their creative design work, and we found that only two participants had well-established routines for revisiting ideas. Through the interviews, we gradually discovered that all participants in fact had some form of existing idea workflows. The participants had however not received any formal training or education in this, and they had each developed their workflows over time through iterations of trial and error. As the study progressed, the participants became more aware of and articulate about these processes, and how they might potentially be improved or transformed in the future.

The lack of structured approaches to revisiting ideas is interesting in light of several strands of research that demonstrate the potential value of idea recollection. In terms of *idea management in design*, Inie and Dalsgaard (2020) demonstrated how revisiting old ideas can be of value to designers, both in terms of informing ongoing projects, yielding ideas for new projects and simply just feel meaningful as a practice of reminiscing. In a broader perspective, the extensive literature on *spaced repetition* (Ausubel & Mohamed, 1965) has shown how revisiting information with regular intervals can greatly enhance subjects' ability to recall it. This insight can be interpreted in relation to the notion of repertoire in design. As argued by Schön (1979) and many subsequent design researchers informed by his work (e.g. Dalsgaard, 2014; Gedenryd, 1998), design competence relies in part on practitioners' ability to draw on their repertoires and see how current design problems have common traits with prior ones. Given that spaced repetition enhances the ability to recall such prior information, we can be hypothesize that forms of spaced repetition of design ideas can enhance designers' abilities to bring their repertoire into play. Conversely, designers' ability to exploit their repertoire may suffer if they do not have routines for revisiting ideas. With this in mind, the finding that the study participants' work practices did not include structured routines for revisiting ideas was unanticipated. While the strong solution-oriented

focus discussed above can serve to explain this pattern, there is an untapped potential for better supporting recollection of design ideas.

As the interviews were conducted over the course of four weeks, we slowly discovered that even though most participants reported a lack of structure in their idea workflows, they in fact all had some form of existing idea workflows in place. As the study progressed, the participants became more aware of and articulate about these processes, and how they could be improved or transformed in the future. At the end of the study, most participants had developed clear views about needs and requirements for future systems to integrate into their work practices. They stated that their participation in the study had helped them reflect on and develop opinions about what they would like to see in future idea management systems. The diversity in their current practices and routines shone through in their wishes for future systems to support their idea workflows, and they had very different views on what would constitute the 'ideal tool for idea collection and development,' for instance as seen in section 4.3.2.

## 5 *Conclusion*

This exploratory, longitudinal study offers insight into professional designers' idea workflows and practices. Using a technology probe, Recollect, which is as a custom-made idea management web application for smartphones, this study has shed light on how ten professional designers conceptualize, capture, manage, develop, and share ideas as examined over a month-long period. The participating designers were prompted to articulate and develop their understanding of their current practices around idea management and development, specifically relating to their use of tools. The first central insight on current idea practices indicated that the designers agree on the perception of ideas as "solutions", which they work rapidly toward implementing, and that engagement in active recollection or revisiting of ideas is rare. However, the second insight highlights that while the designers tend to agree on what constitutes a design idea, they have completely distinct practices for working with these ideas. On a different level of abstraction, a third insight is how professional designers generally claim to find value in revisiting ideas in various ways, which is in contrast to the fourth insight that brings to the fore the seeming lack of structured approaches for actually accomplishing this in practice. Pairwise, these four insights form two critical tensions in the ways that current idea management is conducted over longer periods of time by professional designers.

We see two main strands of implications of this research. From a research perspective, the tensions we have identified call for additional research on the complexity of real-life professional idea management practices. In particular, we find it relevant to further examine if and why there are patterns in the

very diverse and seemingly idiosyncratic ways in which designers capture and manage ideas. This could shed further light on whether these differences are a result of individual preferences, organizational frames and structures, the affordances, and constraints of tools, and/or other factors. From a practice-based perspective, these findings have implications for designers' routines and workflows and the development of new tools to support idea management. The most striking finding in our view is that there seems to be an untapped potential in revisiting ideas. Regardless of their diverse workflows, all participants found great value in incorporating new routines for revisiting ideas and expressed in the interviews that they would continue to do so beyond the study. In terms of implications for designing new tools, the diversity of practices, preferences and use of existing tools indicate that modular and customizable tools, as well as means for better integrating existing tools, would be a more apt approach than one-size-fits-all solutions. However, better means for revisiting ideas is a general principle could be integrated into many types of tools, and this to us would be a promising avenue for future systems. While we have focused here on designers, we speculate that our findings can also serve as a basis for new product development initiatives for a broader range of creative professionals, who rely on easy and efficient idea management practices in their daily work.

## *Declaration of competing interest*

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## *Data availability*

The data that has been used is confidential.

## *Acknowledgements*

This project is funded by Independent Research Fund Denmark 0132-00111B, Managing Ideas in Creative Work.

## *References*

Arthur Koestler. (2020). *The act of creation*. University of California Press.

Ausubel, D. P., & Mohamed, Y. (1965). The effect of spaced repetition on meaningful retention. *The Journal of General Psychology, 73*(1), 147–150.

Ball, L., & Christensen, B. T. (2018). Designing in the wild. *Design Studies, 57*(1–8).

Biskjaer, M. M., Dalsgaard, P., & Kim, H. (2017). Understanding creativity methods in design. In *Proceedings of the 2017 conference on designing interactive systems* (pp. 839–851).

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101.

Cardoso, C. C., & Badke-Schaub, P. (2009). Give design a break? The role of incubation periods during idea generation. In *DS 58-2: Proceedings of ICED 09,*

*the 17th international conference on engineering design, 2* (pp. 383–394). Palo Alto, CA, USA: Design Theory and Research Methodology, 24.-27.*08*. 2009.
Carroll, R. (2018). *The bullet journal method: Track the past, order the present, design the future*. Penguin.
Casakin, H., & Levy, S. (2020). Ideation and design ability as antecedents for design expertise. *Creativity Research Journal, 32*(4), 333–343.
Catmull., & Wallace, A. (Eds.). (2014). *Creativity, Inc: Overcoming the unseen forces that stand in the way of true inspiration*. Random House.
Christensen, B. T., & Ball, L. J. (2014). *Studying design cognition in the real world using the 'in vivo'methodology*. The Routledge Companion to Design Research.
Christiaans, H. H. C. M. (2002). Creativity as a design criterion. *Communication Research Journal, 14*(1), 41–54.
Chulvi, V., Gonźalez-Cruz, María C., Mulet, E., & Aguilar-Zambrano, J. (2013). Influence of the type of idea-generation method on the creativity of solutions. *Research in Engineering Design, 24*(1), 33–41.
Coughlan, T., & Johnson, P. (2008). Idea management in creative lives. In *CHI'08 extended abstracts on Human factors in computing systems* (pp. 3081–3086).
Dalsgaard, P. (2014). Pragmatism and design thinking. *International Journal of Design, 8*(1).
Davies, R., & J Talbot, R. (1987). Experiencing ideas: Identity, insight and the imago. *Design Studies, 8*(1), 17–25.
Dorst, K., & Cross, N. (2001). Creativity in the design process: Co-evolution of problem–solution. *Design Studies, 22*(5), 425–437.
Finke, R. A., Ward, T. B., & Smith, S. M. (1992). *Creative cognition: Theory, research, and applications*.
Gedenryd, H. (1998). *How designers work-making sense of authentic cognitive activities*. Lund University.
Goncalves, M., Cardoso, C., & Badke-Schaub, P. (2014). What inspires designers? Preferences on inspirational approaches during idea generation. *Design Studies, 35*(1), 29–53.
Guilford, J. P. (1956). The structure of intellect. *Psychological Bulletin, 53*(4), 267.
Haugen Askland, H., Ostwald, M., & Williams, A. (2010). Changing conceptualisations of creativity in design. In *Proceedings of the 1st DESIRE network conference on creativity and innovation in design* (pp. 4–11).
Hollan, J., Hutchins, E., & Kirsh, D. (2000). Distributed cognition: Toward a new foundation for human-computer interaction research. *ACM Transactions on Computer-Human Interaction, 7*(2), 174–196.
Howard, T. J., Culley, S. J., & Dekoninck, E. (2008). Describing the creative design process by the integration of engineering design and cognitive psychology literature. *Design Studies, 29*(2), 160–180.
Hutchinson, H., Mackay, W., Westerlund, B., Bederson, B. B., Allison, D., Plaisant, C., et al. (2003). Technology probes: Inspiring design for and with families. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 17–24).
Inie, N., & Dalsgaard, P. (2017a). How interaction designers use tools to capture, manage, and collaborate on ideas. In *Proceedings of the 2017 CHI conference extended abstracts on human factors in computing systems* (pp. 2668–2675).
Inie, N., & Dalsgaard, P. (2017b). A typology of design ideas. In *Proceedings of the 2017 ACM SIGCHI conference on creativity and cognition* (pp. 393–406).
Inie, N., & Dalsgaard, P. (2020). How interaction designers use tools to manage ideas. *ACM Transactions on Computer-Human Interaction, 27*(2), 1–26.

Inie, N., Dow, S., & Dalsgaard, P. (2018). Supporting reflective use of design idea archives using email. In *Proceedings of the 10th Nordic conference on human-computer interaction* (pp. 726–730).
Inie, N., Frich, J. F., & Dalsgaard, P. (2022). How researchers manage ideas. *Creativity and Cognition* 83–96.
Kaufman, J. C., Plucker, J. A., & Baer, J. (2008)*Essentials of creativity assessment, 53*. John Wiley & Sons.
Kocienda, K. (2018). *Creative selection: Inside Apple's design process during the golden age of Steve Jobs*. Pan Macmillan.
Lou Maher, M., & Poon, J. (1996). Modeling design exploration as co-evolution. *Computer-Aided Civil and Infrastructure Engineering, 11*(3), 195–209.
Löwgren, Jonas, & Stolterman, E. (2004). *Thoughtful interaction design: A design perspective n information technology*. Mit Press.
M Amabile, T. (1988). A model of creativity and innovation in organizations. *Research in Organizational Behavior, 10*(1), 123–167.
M Amabile, T., Conti, R., Coon, H., Lazenby, J., & Herron, M. (1996). Assessing the work environment for creativity. *Academy of Management Journal, 39*(5), 1154–1184.
McGrath, J. E. (1995). Methodology matters: Doing research in the behavioral and social sciences. In *Readings in human–computer interaction* (pp. 152–169). Elsevier.
Na Sio, U., & Ormerod, T. C. (2009). Does incubation enhance problem solving? A meta-analytic review. *Psychological Bulletin, 135*(1), 94.
Newen, A., De Bruin, L., & Gallagher, S. (2018). *The Oxford handbook of 4E cognition*. Oxford University Press.
Oleynick, V. C., Thrash, T. M., LeFew, M. C., Moldovan, E. G., & Kieffaber, P. D. (2014). The scientific study of inspiration in the creative process: Challenges and opportunities. *Frontiers in Human Neuroscience, 8*, 436.
Plucker, J. A., Beghetto, R. A., & Gayle, T. D. (2004). Why isn't creativity more important to educational psychologists? Potentials, pitfalls, and future directions in creativity research. *Educational Psychologist, 39*(2), 83–96.
Puryear, J. S., & Lamb, K. N. (2020). Defining creativity: How far have we come since plucker, beghetto, and dow? *Creativity Research Journal, 32*(3), 206–214.
Rogers, Y., Sharp, H., & Preece, J. (2002). *Interaction design: Beyond humancomputer interaction*. Jon Wiley & Sons. *INC*.
Schon, D. A. (1979). *The reflective practitioner*. New York.
Sharmin, M., Bailey, B. P., Cole, C., & Hamilton, K. (2009). Understanding knowledge management practices for early design activity and its implications for reuse. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 2367–2376).
Smith, S. M., & Ward, T. B. (2012). *Cognition and the creation of ideas*. Oxford Handbook of Thinking and Reasoning.
Sosa, R. (2019). Accretion theory of ideation: Evaluation regimes for ideation stages. *Design Science, 5*, e23.
Terry, G., Hayfield, N., Clarke, V., & Braun, V. (2017). Thematic analysis. *The SAGE Handbook of Qualitative Research in Psychology, 2*, 17–37.
Tsenn, J., Atilola, O., McAdams, D. A., & Linsey, J. S. (2014). The effects of time and incubation on design concept generation. *Design Studies, 35*(5), 500–526.
Vargas Hernandez, N., Shah, J. J., & Smith, S. M. (2010). Understanding design ideation mechanisms through multilevel aligned empirical studies. *Design Studies, 31*(4), 382–410.